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Green Port Transformation in Renewable Energy and Sustainable Water Port Management: Strategies for Climate-Resilient Maritime Infrastructure

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ARTICLE INFO	ABSTRACT
Submission: August 23, 2026 Revised: September 04, 2026 Accepted: September 21, 2026 Available Online: October 01, 2026 Keywords: Greenports, renewable energy, sustainable water management, maritime sustainability, port decarbonization, clean energy, climate-resilient infrastructure, marine environmental management Corresponding author: Captain Muhammad Shahid Email: mshashimi27@yahoo.com	Ports play a critical role in global trade infrastructure, however they are also a significant source of GHG emissions, energy use as well as marine environmental damage. The shift towards “green ports” has become a strategy to lessen the ecological footprint and enhance the efficiency of port operations by incorporating renewable energy sources, sustainable water management, and climate-resilient infrastructure. The role of traditional ports in becoming sustainable, maritime hubs will be discussed and the implementation of renewable energy technologies, including solar photovoltaic (PV) systems, offshore wind, shore-side electricity and clean energy storage, will be outlined. Furthermore, it examines the management of water port in sustainable use of wastewater treatment, ballast water control, rain water harvesting, desalination technologies, as well as protection of marine ecosystems. The study pinpoints technological innovations, policy frameworks and management approaches to developing a low carbon port. The study examines global green port initiatives and sustainable practices to gain insights into the challenges and opportunities of implementing sustainable port operations. Results highlight the need for holistic approaches to energy and water-use management for creating resilient ports to support sustainable shipping and sustainable goals.

Introduction

Ports serve as crucial nodes where sea-based transportation systems connect to land-based ones, and are key contributors to global maritime transportation, which is integral to international trade, economic growth, and supply chain dynamics. But, the port's expansion has brought a number of environmental pressures, such as greenhouse gas emissions, air pollution, energy consumption, waste water and coastal ecosystems. Conventional ports largely rely on fossil fuel-based energy sources to handle cargo, to operate terminals, to serve vessels and to transport cargo, and they are key contributors to the environmental footprint of the maritime sector (Notteboom et al., 2022). With the rising effects of climate change and the international pledges for

sustainability, the port authorities around the world have been encouraged to change their operational models to green and environmentally responsible ones.

Green port transformation has become a holistic approach to minimize environmental impacts while ensuring efficient maritime operations. Green ports integrate technologies, renewable energy resources, environmental monitoring systems, and management strategies to lower carbon emissions and enhance environmental outcomes. The shift to low carbon ports is in line with the global climate targets and will help achieve sustainable development goals by encouraging the adoption of cleaner energy sources, better resource management, and adaptation to climate change in port facilities (Ugwale et al., 2022). The latest research highlights the need for a holistic strategy to promote sustainable port development, encompassing technological innovation, operational optimization, and efficient environmental management.

One of the main enablers towards sustainable port transformation is renewable energy integration. Ports have significant potential to implement renewable energy solutions, owing to their high land availability, coastal position and energy need throughout the year. Port installations of solar PV power, wind energy, renewable shore-side power and storage system solutions can help to lower fossil fuel reliance and emissions from port operations. In addition, smart energy management systems can help ports manage their energy use more efficiently by coordinating energy generation from renewables, energy storage, and industrial energy demand (Yang et al., 2025). Renewable energy technologies also help to enhance energy resilience in the face of higher fuel price variability and climate variability and uncertainty.

Sustainable water management has also become a key part of the sustainable development of modern green ports along with the energy transformation. Ports are situated in highly sensitive coastal areas where the water quality is influenced by industrial discharge, vessel operations, stormwater runoff and accidental pollution. Wastewater treatment, pollution prevention, ballast water control, water recycling, and nature-based solutions are crucial for the protection of marine ecosystems and environmental sustainability in the long-term (Garcia-Herrero et al., 2023). In addition, advanced water treatment technologies and renewable (RE) desalination systems are available for ports in water-stressed coastal areas that can enhance freshwater availability while minimizing environmental impacts (Al-Karaghoul & Kazmerski, 2013).

Renewable energy and sustainable water management are some of the first steps towards a more integrated approach to port sustainability. Smart technologies, automated monitoring systems, optimizing through A.I. and decision making based on data are some of the resources that are increasingly used in Green ports to enhance resource efficiency and environmental performance. These strategies enable port authorities to track emissions, manage energy use, control the quality of the water, and employ preventive environmental measures (Yang et al., 2025). Yet, the process of green port transformation is not easy as there are many barriers, such as high investment, technology barriers, regulatory disparity, and even coordination among various marine stakeholders.

To build sustainable port infrastructure, cooperation among governments, port authorities, shipping companies, energy providers and the local community is needed. Policy frameworks, financial mechanisms and international environmental regulations are important drivers of increased uptake of renewable energy and sustainable use of resources. While there have been efforts by several major ports to establish green port programs, there are still challenges in developing sustainable programs in many developing ports because of inadequate infrastructure and investment.

For this reason, this research looks at the change of ports due to the use of renewable energies and sustainable water management measures. The research investigates the new technologies, environmental advantages, project challenges, and potential opportunities for green port development. This study analyzes the correlation between integrating clean energy and sustainable water practices and the role of ports in future maritime transportation systems to help understand how the port sector can become climate-resilient and environmentally responsible.

Literature Review

Evolution of Green Port Concepts and Sustainable Maritime Development

Given the high growth rate of global maritime traffic, the "sustainability" and "low carbon" aspects of port management have grown in importance, with a shift from traditional port management towards the sustainable and low carbon management. While the economic development and growth and efficient handling of cargo and infrastructure development are the main concerns of the ports traditionally, the high level of concern on climate change, air pollution and degradation of marine ecosystem has brought the attention to climate friendly development of ports. The green port concept has become a strategic approach by

incorporating environmental considerations with operational efficiency in the use of cleaner energy, emission reduction measures, sustainable use of resources and ecological conservation (Notteboom et al., 2022).

Recent studies have underscored the need for a multidisciplinary approach to transform the green port, which encompasses technological innovation, environmental governance, and collaboration with stakeholders. The concept of "sustainable ports" is not simply about handling cargo; it also encompasses the reduction of carbon emissions, enhancing energy efficiency, optimizing waste resources, and safeguarding local marine ecosystems (Hossain et al., 2022). Adoption of green port initiatives has been spreading across the world, with initiatives like the World Ports Climate Initiative, EcoPorts and region-wide sustainability frameworks guiding port operators to adopt measurable environmental performance indicators.

There is literature that suggests that successful green port development relies on developing a long-term strategy that incorporates sustainability as a key element of the port's development as opposed to sustainability being an activity that is performed as a separate operation. Adaptable infrastructure, renewable energy, and environmental monitoring systems have become recognized as key elements of future maritime infrastructure planning by port authorities, as shown by Pourmohammad-Zia and van Koningsveld (2025). It reflects a wider trend towards ports that are resilient to climate hazards that can operate when environmental conditions are worsening, including sea-level rise, extreme weather events, and the changes in coastal ecosystems.

Renewable Energy Integration in Green Port Transformation

The energy needs of maritime operations make renewable energy a key component of green port transformation. Ports use a lot of electricity and fuel for cargo cranes, terminal equipment, warehouses, warehouse lighting, and port support services. Conventional energy dependence plays a large role in both greenhouse gas emissions and energy expenditures, necessitating the availability of greener and more efficient sources of energy.

Due to the fact that a lot of rooftops, warehouses, parking spaces and unused land is available in a port, solar photovoltaic technology has been one of the most widely used renewable energy technologies in port area. Resence studies show that solar PV systems can be utilized to power end-of-life activities and lower reliance on energy sources from the grid, excluding fossil fuels. Likewise, wind energy systems in coastal areas offer further possibilities for renewable electricity generation, as there is favorable wind in coastal areas (Acciaro et al., 2023).

Shore-side electricity, or "cold ironing" as it is also referred to, is another key element of renewable energy that can be used when vessels are moored to the local electricity grid, aiding them in switching off their auxiliary diesel engines. This will lower emissions of carbon dioxide, NO_x, SO_x and PM within port areas, resulting in better air quality for port workers and local communities. Studies have shown that the electrification of port operations, coupled with the use of renewable energy, can play a significant role in minimizing the environmental footprint resulting from port and vessel activities (Yang et al., 2025).

The integration of renewable energy has been enhanced by the development of various energy storage technologies and the implementation of smart energy management systems. Battery storage systems can help ports to regulate the variability of renewable energy generation and provide consistent power supply during peak loads. Energy management systems in smart ports can help minimize energy use and carbon emissions by optimizing electricity distribution, equipment scheduling and processes (Yang et al., 2025). There are, however, challenges in terms of cost of investment, infrastructure compatibility, cybersecurity, and a uniform system of energy management.

Sustainable Water Management in Port Environments

Another critical aspect of a sustainable port is water management since there is a direct linkage between port activities and marine ecosystems. Coastal areas are where ports are located, and industrial discharge, vessel operations, stormwater runoff and accidental spills can adversely impact water quality in the area. The sustainable management of water sources is designed to reduce harmful pollutants, save freshwater and preserve aquatic biodiversity.

Wastewater treatment is one of the most important issues in the management of water for sustainability in ports. Wastewater is produced at the port from maintenance tasks, vessel services, industrial activities, and domestic use. Advanced treatment technologies such as biological treatment systems, membrane filtration and constructed wetlands have been observed as environmentally friendly approaches to water quality improvement prior to discharge. Garcia-Herrero et al. (2023) noted that NbWTS can offer environmental benefits, decrease pollutant loads, and contribute to biodiversity conservation.

There is also a strong environmental concern for ballast water management as ships can carry aquatic invasive species in their ballast water discharge. International standards have been developed to ensure that biological contamination can be minimized and damage to marine systems avoided through effective treatment systems. Sustainable Ballast Water Management practices help to preserve the ecological health and support responsible maritime operations.

The concept of water conservation has gained traction in sustainable port planning, such as rainwater harvesting, water recycling, and desalination using renewable energy. The following strategies apply specially to ports in areas where freshwater is scarce. The use of water efficient technologies will help ports to lessen their dependence on freshwater and boost resilient response to the water challenges of climate change (Al-Karaghoul & Kazmerski, 2013).

Smart Technologies Supporting Sustainable Port Operations

Innovation has become vital in the field of environmental performance of ports. Smart port technologies, such as artificial intelligence, Internet of Things (IoT) systems, digital twins and automated monitoring platforms, offer ways to optimize energy use, manage resources and take better decisions about the environment.

Digital twin technology has become a spotlight in the field of sustainable port management. The digital twin is a virtual model of the port infrastructure and the processes of port operations, which can be used to track the performance, analyze the environmental impacts, and test potential improvements prior to their implementation. Klar et al. (2023) proposed digital twins to be used for energy optimization, resource allocation and multi-stakeholder collaboration in complex port systems.

Environmental monitoring technologies also help to ensure sustainable water and energy use by providing data on emissions, water quality, energy consumption and efficiency in real-time. The systems enable port managers to recognize potential environmental hazards and take preventive measures. Smart technologies can be linked with the use of renewable energy and efficient use of resources and offer possibilities to develop more adaptive and efficient port ecosystems.

Challenges and Future Directions of Green Port Transformation

A number of challenges remain to facilitate the full and rapid adoption of green ports. Despite the advances made in green ports, there are still some obstacles to be overcome before green ports can be widespread. Initial investment needs are high, there is limited access to renewable energy infrastructure, the technological complexity is still present and there are regulatory differences between countries. For developing countries, further limitations can exist because of financial constraints and lack of technical capacity.

The literature shows that government, port, shipping company, technology provider and local community should be more closely coordinated in the future development of green ports. Long-term environmental targets require integrated planning approaches of the combination of renewable energy systems, sustainable water management and digital technologies. Further research is needed to create cost-effective solutions, policy frameworks, and assess the multi-facet environmental and economic impact of a sustainable port transformation.

Generally, the current literature shows that the transformation of the port cannot be restricted to the field of emissions reduction, but is a complete process that contains elements of energy transition, water saving, technological innovation and adaptation to climate change. The use of renewable energy and sustainable water management are key examples of a foundation for developing resilient maritime infrastructure for sustainable global trade.

Methodology

Research Design

The methodology used in this study is qualitative research, which is carried out based on a comprehensive review and analysis of the existing academic literature and industry reports, sustainability frameworks and international green port initiatives. The research method focuses on how the conventional ports are changing to sustainable maritime infrastructures by integrating renewable energy and sustainable water management. A systematic evaluation framework is used to detect key technological innovations, environmental measures, operational issues and future prospects for the green port transformation.

The methodology draws together the concepts of sustainable infrastructure development, renewable energy transition, marine environmental management and climate-resilient port planning. The analysis covers renewable energy technologies such as solar

PV, wind power, shore-side electricity, hydrogen technologies and energy storage solutions and their role in the reduction of emissions and improvement in energy efficiency. Likewise, sustainable water management solutions, such as wastewater treatment, ballast water control, water recycling, stormwater management and nature-based solutions are evaluated based on their environmental advantages and their suitability in port settings.

Data collection and data sources

This research used peer-reviewed scientific studies, international sustainability reports, maritime policy papers and technical research on green ports and sustainable infrastructure development. The literature review is mostly devoted to the contributions of works published from 2020 to 2026, as this period reflects the latest advancements in the field of renewable energy technologies, smart port development, and environmental management practices.

Academic databases and research platforms were searched to find suitable academic research on port decarbonization, renewable energy integration, and sustainable water management. Additional information collected from international organizations in the field of Climate Action, Renewable Energy Development and Maritime Sustainability. The transition to renewable energy highlights the importance of solar, wind, and other green energy resources in cutting reliance on fossil fuels and helping develop low-carbon infrastructure (United Nations, 2025).

Analytical Framework

A sustainability assessment framework has been created to assess the green port transformation from three angles:

Renewable Energy Integration Assessment

This dimension will assess implementation and potential contribution of clean energy technologies in port operation. Assessment indicators are:

- The amount of renewable energy power produced.
- The decreased reliance on fossil fuels. Less reliance on fossil fuels.
- The potential for reducing carbon emission.
- Energy storage integration
- Increasing the use of electrical power for port machinery. Port equipment electrification.
- Shore-side electricity availability

It takes into account the potential for renewable energy systems to be integrated into the port operations without compromising environmental performance. The previous studies have shown that energy management systems and renewable energy can help to save energy and greenhouse gas emissions while cutting the operational costs in smart ports (Yang et al., 2025).

Sustainable Water Management Assessment

The second dimension looks at efforts made by ports to safeguard marine environment and increase water resource efficiency. Assessment indicators are:

- Wastewater treatment efficiency
- Residuals from ballast water management practices
- Water recycling systems
- Stormwater pollution control
- Marine ecosystem protection
- Freshwater conservation strategies

The management of water resources in sustainable ways is becoming more crucial with the increasing environmental stress and climate-related water issues on coastal infrastructure. Constructed wetlands (CWs) have been shown to have the potential to treat pollutants and offer ecological benefits (Garcia-Herrero et al., 2023).

Smart and Resilient Port Management Assessment

The third dimension looks at the contribution of cutting edge technologies and management systems to drive environmental performance. The following is assessed:

- Artificial intelligence-based monitoring
- Internet of Things (IoT) applications
- Digital environmental monitoring
- Automated energy management
- Climate adaptation planning
- Data-driven decision-making

Savvy technologies enable port authorities to track resource use, optimize operations and enhance their environmental decision making. The study on smart port energy management shows integrated monitoring systems can be used to help reduce emissions and improve the operational efficiency of energy management (Yang et al., 2025).

Case Study Selection Approach

Selected international ports with a good track record of sustainability initiatives are analyzed. Case examples are chosen based on the availability of documented environmental programs, renewable energy projects and sustainable water management practices. The chosen cases are based on various geographical areas and operational environments to offer a wider sense of the various green port transformation strategies.

The case study analysis is conducted on:

- Adoption strategies for renewable energy
- Carbon reduction initiatives
- Water conservation programs
- Environmental governance frameworks
- Technological implementation challenges

The comparative approach allows to outline successful practices and barriers to sustainable port development.

Data Analysis Method

Thematic analysis was used to analyze the information collected. The major themes were identified which are relevant to the study the renewable energy transition, water sustainability, technological innovation, policy support, and implementation barriers. The analysis reveals the relationships between sustainable energy use and environmental improvements in port systems.

Results are analysed using principles of Sustainable Development and Climate Resilient Infrastructure planning. The methodology also recognizes that a transition to a green footprint of the port cannot be made through taking individual environmental measures. Considering the interdependence between water and energy and the environment, renewable energy and water management strategies are seen as being interconnected to sustainable port ecosystems.

Research Limitations

This study's findings were limited in a number of ways. The research relies mostly on secondary data sources, which means that there can be differences in data availability, data reporting, sustainability assessment criteria, etc. between ports. Furthermore, the extent to which the technologies are being adopted is different across the regions, and there are different environmental policies that can affect the assessment of green port projects. Moreover, the extent of technology uptake varies across regions, along with the various environmental policies that are in place; this can affect the comparison of green port projects.

They may be combined with quantitative modelling, lifecycle assessment approaches, and data gathering from port authorities in the future to achieve more comprehensive evaluation of environmental and economic results related to green port transformation.

Results and case study analysis

The analysis of the green port initiatives around the world shows that the process of sustainable port transformation is developing in three interrelated directions: smart environmental governance, sustainable water use and renewable energy use. The leading ports have advanced beyond the implementation of individual environmental projects and are now creating an integrated approach to sustainability – one that incorporates the use of clean energy infrastructure, emission reduction initiatives, protection of resources and digital monitoring systems. Through electrification, alternative fuels, renewable energy systems and

energy efficiency improvements, major ports such as Singapore, Rotterdam, LA-Long Beach, and other ports around the world have established defined decarbonisation paths for their operations (Chen et al., 2026; Notteboom et al., 2022).

The comparative assessment showed that the successful green ports tend to have certain common features such as advanced monitoring technologies, good stakeholder cooperation, investment in renewable infrastructure, and strong environmental policies. Rather, integrating renewable sources into energy production is a significant aspect of sustainability initiatives, and managing water resources is a key part of initiatives aimed at reducing marine pollution and safeguarding coastal systems.

Table 1: Renewable Energy and Sustainability Practices in Selected Global Green Ports

Port	Region	Renewable Energy Initiatives	Sustainable Practices	Management	Key Outcomes
Port of Rotterdam	Netherlands	Offshore wind integration, renewable electricity, hydrogen projects, smart energy systems	Industrial energy transition, emission monitoring, circular resource management		Reduced dependence on fossil fuels and development of clean energy infrastructure
Port of Singapore	Singapore	Solar energy systems, clean energy planning, alternative fuel research	Green shipping corridors, digital monitoring, environmental management programs		Improved energy efficiency and support for low-carbon maritime operations
Port of Los Angeles	USA	Shore-side electricity, renewable energy projects, electrification of port equipment	Clean truck programs, stormwater management, emission control systems		Reduced vessel emissions and improved air quality near port communities
Port of Antwerp-Bruges	Belgium	Wind turbines, solar installations, renewable hydrogen development	Circular economy programs and industrial sustainability initiatives		Increased renewable energy capacity and cleaner industrial operations

Renewable Energy Integration Results

The findings showed that the use of renewables is one of the most viable mitigation measures for carbon emission caused by port operations. However, because of its non-invasive nature, electricity generation from solar PV is being placed on warehouses, terminal buildings and unused parts of port sites. In addition to this, the wind energy potential is favorable at coastal ports because of the favorable maritime conditions. Research works on renewable energy in ports have shown that solar, wind, hydrogen and storage technologies can help to reduce emissions and enhance energy security in port areas. Research about renewable energy in ports has highlighted the ability of technologies such as solar, wind, hydrogen and storage to support emission reduction and improve energy security in port areas.

The Port of Rotterdam is a good example of transition to renewable energy through renewable electricity systems, hydrogen projects and industrial energy networks. The port's sustainability plan is dedicated to achieving emissions reduction from one of Europe's most significant industrial clusters without impacting the economic activities of the port.

Likewise, Singapore has introduced the concept of renewable energy planning as part of decarbonization efforts for maritime transportation. The Green and Digital Shipping Corridor (GDSC) between Singapore, Los Angeles and Long Beach is a prime example of how large ports are collaborating to research alternative fuels, digital solutions and help save energy.

The Port of Los Angeles has already shown how critical the electrification of vessels is by implementing shore-side electricity systems, which provide a connection to external power sources for docked vessels to power auxiliary diesel engines. This way, nitrogen oxides and particulate matter released during vessels' movement at berth are reduced.

Sustainable Water Management Results

The assessment of sustainable water use shows that green ports are increasingly aware of the importance of water management for their environmental performance. Pollutants to the port can come from vessel operations, stormwater runoff, industrial activities, and wastewater discharge. Hence, the advanced systems of treatment of polluted waters, prevention of polluted water and ecological restoration in modern ports are being carried out.

Some of the key sustainability actions identified are:

- Wastewater treatment facilities
- Collect and filter storm water.
- Ballast water treatment technologies
- Water recycling systems
- Marine habitat restoration programs.

An example of integrated water protection strategies is the Port of Los Angeles' stormwater management programs that help prevent pollutants from entering the marine environment. (lpj.gov.my)

Constructed wetlands and ecological treatment systems are also being considered as part of nature-based water management solutions as they offer both biodiversity conservation and pollution control. These methods are especially useful for ports that are aiming to achieve long-term environmental resilience (Garcia-Herrero et al., 2023).

Table 2: Sustainable Water Management Practices in Green Ports

Water Management Strategy	Application in Ports	Environmental Benefits
Wastewater Treatment	Treatment of industrial and operational wastewater before discharge	Reduces marine pollution
Ballast Water Management	Treatment of ship ballast discharge	Prevents invasive species introduction
Stormwater Control	Collection and filtration of contaminated runoff	Protects coastal water quality
Water Recycling	Reuse of treated water for industrial activities	Reduces freshwater demand
Nature-Based Solutions	Wetlands and ecological treatment systems	Supports biodiversity and climate adaptation

Role of Smart Technologies in Green Port Performance

The conclusion highlights the fact that digital technologies are emerging as an essential tool to enhance environmental management in ports. Authorities can use smart monitoring systems to gather real-time data on energy usage, emissions, water quality and performance.

AI, the Internet of Things (IoT) and digital twin technologies assist in predictive management and help spot inefficiencies and optimize resource allocation. These equipments help ports to manage energy distribution and optimize it, keep track of environmental conditions and facilitate the coordination between shipping companies, the port operators and regulatory agencies.

Studies on green port development over recent years have highlighted the necessity to take into account operational data, renewable energy availability, and infrastructure conditions when assessing the sustainability performance of ports (Chen et al., 2026).

Comparative Sustainability Assessment

Table 3: Comparative Evaluation of Green Port Transformation Dimensions

Sustainability Dimension	Rotterdam	Singapore	Los Angeles
Renewable Energy Adoption	High – wind, hydrogen, renewable electricity	High – solar and clean energy initiatives	Moderate–High – electrification and renewable projects
Emission Reduction Strategies	Hydrogen development, energy transition	Green shipping corridors and alternative fuels	Shore power and clean transportation programs
Water Management	Industrial water management and pollution control	Marine environmental protection programs	Stormwater management and coastal protection
Smart Port Development	Advanced energy networks	Digital maritime systems	Automated monitoring and emission tracking

The results indicate that the green transition of ports should go hand in hand with a multi-faceted development of the renewable energy systems, sustainable water management and smart operational technologies. By using renewable energy and water sources in a sustainable way, we reduce the use of fossil fuels and support the health and well-being of marine life and water resource efficiency. Digital technologies have the potential to better support environmental monitoring and support the evidence-based decision making processes.

The case studies show that successful green ports are the result of a long term planning, investment and regulatory process and a cooperation between different maritime stakeholders. However, the level of implementation and the extent of sustainability depends on the different levels of resources, financial capacity and policy frameworks across the regions.

Overall, the results have given confidence that the development of renewable energy and the sustainable use of water can complement each other to make ports resilient and viable for the future of maritime transportation system.

Discussion

The findings of this research reveal that Green Port Transformation is a multi-faceted process, which includes the use of Renewable Energy, Sustainable Water Use, Technological Innovation and Environmental Governance. The conventional ports have always been interested in optimizing their operation and handling of goods, but with an increasing awareness of environmental concerns, the focus has now been on the development of environmentally friendly ports. The findings corroborate earlier studies showing that modern ports should prioritize ecological and operational sustainability in order to mitigate the negative effects of maritime operations on the environment while meeting the challenges of climate change (Notteboom et al., 2022). Moving to green ports necessitates coordinated strategies to tackle the energy consumption, carbon emissions, marine pollution and climate resilience aspects.

Incorporating renewable energy sources is becoming an increasingly important approach to decreasing emissions from port activities. The results show that renewable energy sources like solar power, wind energy, renewable hydrogen, shore-side electricity and energy storage technologies have the potential to play a major role in fossil fuel reduction. Ports use a lot of energy in handling of cargo, as well as in terminal operations, lighting systems and services for the ships that enter into the port's facilities; so alternative energy sources can be used to improve environmental performance, especially when replacing conventional energy sources with renewable ones. Some of the previous works have reported that the integration of renewable energy in port infrastructure with smart energy management systems can optimize the use of electricity, mitigate greenhouse gas emissions and enhance energy security (Acciaro et al., 2023). But the uptake of renewable energy continues to be affected by a number of influencing factors including infrastructure, investment requirements, technology availability and regulatory support. Thus, a long-term sustainability planning and financial frameworks must be developed to enable broad implementation, especially in the developing maritime regions.

The results also highlight the importance of sustainable water management in green port development. The importance of ports links directly to coastal ecosystems and proper water governance is needed to limit marine pollution and biodiversity loss. Wastewater treatment, ballast water management, storm water control, and water recycling technologies offer potential risk reductions to the environment from port activities. Previous studies have shown that NbS such as ecological treatment systems (ETS) and constructed wetlands (CW) can benefit water quality and ecosystem conservation (Garcia-Herrero et al., 2023). Good water management practices can also help to build resilience to flood, freshwater shortage and the effects of a changing coastal environment as part of climate adaptation. However, this will need ongoing cooperation, technical support and monitoring between environmental agencies and port authorities if it is to be effective.

The study also raises the issue of the growing role played by smart technologies in the context of sustainable port operation. Digital monitoring systems, AI applications, Internet of Things (IoT) and digital twin technologies offer great potential to enhance resource management and decision-making around the environment. These technologies can contribute to monitoring energy use, emissions, and water quality in real time and allow for predictive maintenance and optimizing operations. The integrated energy management systems in smart ports have been shown to enhance energy efficiency by optimizing the utilization of renewable energy generation, storage capacity, and the operational demand (Yang et al., 2025). While digital transformation offers many benefits, it also presents challenges in the areas of cybersecurity, data management, infrastructure compatibility, and skills. They will need to be addressed to fully capitalize on the benefits of smart sustainable port systems.

International green port experiences reveal that policy as well as the cooperation among stakeholders are critical to the successful realization of green port transformation. It's up to governments, port authority, shipping companies, energy providers,

technology developers and local communities to set clear sustainability goals and work out a plan for how to achieve them. The transition to cleaner technologies and emissions reduction is a measure encouraged by environmental regulations and international maritime policies, which still have an effect on the sustainability progress of regions, depending on their economic capacity and the different regulatory frameworks. Green port solutions need to be adopted in developing economies with further support in terms of investment, knowledge sharing and technology transfer.

The results indicate that the future development of the green ports must go beyond the environmental projects to the integrated models of sustainability, which involve renewable energy, water management and digital technologies. Contrary to the prevailing research on the energy transition and on environmental protection, the current research takes a more holistic approach, viewing energy, water and ecology in relation to one another within the context of the port. More detailed assessment tools which evaluate environmental, economic and operational impacts at the same time should be developed for future studies. Moreover, life cycle assessments (LCA) of renewable energy infrastructure, hydrogen systems and energy storage technologies should be done to establish long-term sustainability impacts.

Overall, the results of this study confirm the hypothesis that a green transformation of ports is possible only by implementing renewable energy and sustainable water and wastewater management systems. Renewable energy contributes to energy resilience and reduces carbon emissions, while sustainable water management helps protect marine ecosystems and promotes ecological balance. These can go hand-in-hand with smart technologies and good governance structures and provide the groundwork for climate-smart maritime infrastructure. The journey towards sustainable ports demands constant innovation, investment, and collaboration with stakeholders for long-term environmental and operational sustainability.

Conclusion

This is a critical demand to address global climate change, environmental degradation and the growing requirements of maritime operations for traditional ports to shift to green and sustainable infrastructure. This study supports the argument that a holistic approach is needed to create the green port, through a combination of renewable energy, efficient use of water, smart technologies and robust environmental governance. The economic role of ports is no longer the only one they have; they are now considered as elements to sustainable coastal development initiatives.

The results show that renewable energy integration offers great potential to lower GHGs and fossil fuel usage. The use of renewable energy technologies including solar PV, wind energy, shore-side electricity, renewable hydrogen and energy storage can help make ports cleaner and more efficient and resilient. However, for successful implementation, it is necessary to have the right infrastructure, financial investment, supportive policies, and work collaboration among maritime stakeholders.

Water use in a sustainable way is also one of the main areas of the green port transformation. Ports are directly engaged with marine ecosystems, and thus, it is essential to manage wastewater, stormwater runoff, ballast water and freshwater resources effectively to minimize pollution and protect coastal ecosystems. A better water sustainability by using advanced water treatment technologies, recycling systems and NbS can help to conserve the marine ecosystems.

Furthermore, the study highlights the importance of smart technologies for improving the environmental performance and operational efficiency. The opportunities for real-time assessment and optimisation of port activities include AI, IoT systems, digital monitoring platforms and energy management solutions. The technologies enable better decision making, and assist in responding to environmental issues at ports.

Though it has made tremendous progress, there are some implementation challenges, including high costs, technical challenges, regulatory differences, and a lack of sustainability capacity in developing regions. The solution to these challenges must be achieved through good policy frameworks, investment pathways, learning and international collaboration. Overall, the transformation of the port towards greenness is a long process that will need to be a continuous process of innovation and coordination to ensure that ports are environmentally responsible and resilient to climate change.

Recommendations

Port authorities need to create comprehensive sustainability plans by incorporating renewable energy, water-saving measures, emission reduction, and climate adaptation solutions into long-term development plans to speed up the green port transformation. Environmental goals need to be integrated into port management and not seen as standalone projects.

Ports need to further invest in renewable energy, including renewable fuels and renewable energy storage systems, wind energy systems and solar energy generation. Minimisation of emissions from vessels when at dock should also be promoted by the regeneration of shore-side electricity powering. These should be encouraged by the use of funding schemes, incentives and public-private partnerships by governments and financial institutions.

Water resources management should be a key element in the environmental planning for sustainability in ports. The port authorities should implement: Advanced wastewater treatment systems, water recycling programmes, ballast water management technologies and levels of pollution monitoring. Where appropriate, use of NbS to enhance water quality and bolster coastal ecosystem protection should be encouraged.

More and better monitoring, operational efficiency and sustainability assessment should be achieved by scaling up smart technologies. On the other hand, digital platforms, AI systems and the monitoring of the environment in real time can help optimise energy consumption and manage the resources and detect potential environmental risks at port areas.

The effective implementation of green port measures requires improved governmental cooperation, maritime cooperation, shipping industry cooperation, cooperation of energy sector, development and adoption of technology and cooperating with local communities. The areas of international co-operation that are important include technology transfer, capacity building or financial assistance, particularly for developing countries which have weak infrastructure and resources.

Based on the above, it is recommended that further research be conducted on the development of standardised sustainability indicators for green ports in the future. Comprehensive environmental and economic performance of RE systems, sustainable water utilisation and smart technologies should be studied under the concept of LCA and long-term performance evaluation in future. This study can contribute to the evidence-based decision-making process, and can also be utilized in developing sustainable maritime infrastructure in the world.

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